



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

JUNE 2016

CHEMISTRY 9189/1

- 1 (a) (i) gas that obeys the equation, $pV = nRT/AW$; Boyle's law [1]
- (ii) - noble gases are monoatomic, giving them a small ~~molecule~~ ^{particle} size which makes the volume of individual ~~molecules~~ ^{atoms} negligible compared to the total volume of gas; AW [1]
- weak ~~van der Waal's~~ ^{intermolecular} forces between molecules cause ~~molecules~~ ^{particles} not to stick together; AW show comparison [1]

(b) (i) Number of moles of $F_2 = \frac{185}{38} = 4,868$; [1]

$\therefore$ Total number of gas moles $= 4 + 4,868 = 8,9$ [1]

$v = \frac{nRT}{P} = \frac{8,9 \times 8,31 \times 273}{2,5 \times 10^5} = 0,081 \text{ m}^3$ (R) wrong units [1]



(iii) 1. Number of moles of $Li = \frac{23}{7} = 3,3$; ^{3,286} [1]

Number of moles of $F_2 = \frac{1}{2} \times 3,3 = 1,65$; ^{1,65} [1]

Change in number of moles of $F_2 = 4,868 - 1,65 = 3,218$ [1]

Total number of gas moles remaining $= 4 + 3,218 = 7,218$ [1]

2. Change in pressure $= 2,5 \times 10^5 - \left(\frac{7,218 \times 8,31 \times 273}{0,081} \right)$; ⁴⁷⁸⁵⁹ [1]

$= 45\,000 \text{ Pa} / 0,45 \text{ atm}$; ^{45312,85 Pa} [Total: 12]

2 (a) (i) $1s^2 \quad 2s^2 \quad 2p^6 \quad 3s^2 \quad 3p^6$; [1]

(ii) D – second ionisation enthalpy of calcium; [1]

E – first electron affinity of sulphur; [1]

(iii) electron nearer to the nucleus/electron removed from a positive species ^{in effective nuclear charge} [1]

(iv) the negative s^- ion repels the added electron; [1]

(b) (i) no change; equal number of gas moles on reactants and products; [1]

{ 1 }

{ 1 }

(ii) 1. Moles of $I_2 = 0,023$; [1]

2. moles of $HI = 0,172$; [1]

$$K_c = \frac{[H_2][I_2]}{[HI]^2}; \quad [1]$$

$$\frac{(0,023)^2}{(0,172)^2} = \quad [1]$$

$$1,79 \times 10^{-2}; \quad [1]$$

3 (a) (i) - on addition of acid $HCO_3^- + H^+ \rightleftharpoons H_2CO_3$; [1]

- on addition of base $H_2CO_3 + OH^- \rightarrow HCO_3^- + H_2O$; [1]

(ii) $[H^+] = 3,89 \times 10^{-8} \text{ mol dm}^{-3}$; $pH = pK_a + \log \left(\frac{\text{salt}}{\text{acid}} \right)$ [1]

$$K_a = \frac{(3,89 \times 10^{-8})(2,5 \times 10^{-2})}{(1,25 \times 10^{-2})} = 7,78 \times 10^{-8} \text{ mol dm}^{-3}; \quad pK_a = 7,41 + \log \left(\frac{\text{acid}}{\text{salt}} \right) [1]$$

$$K_a = 10^{-6,109} = 8,12 \times 10^{-7} \quad [1]$$

(b) (i) $pH = pK_a + \log \left(\frac{\text{salt}}{\text{acid}} \right);$

$$= -\log(1,74 \times 10^{-5}) + \log \left(\frac{0,10}{0,15} \right)$$

$$= 4,759 - 0,17609; \quad [1]$$

$$= 4,93; \quad 4,58 \quad [1]$$

(ii) Number of moles of H^+ added $= 10 \times 10^{-3} \times 0,1 = 0,001$ 0,01 [1]

Number of ethanoic acid after addition $= 0,15 + 0,01 = 0,16$ 0,16 [1]

Number of ethanoate after addition $= 0,10 - 0,01 = 0,09$ [1]

$$pH = -\log 1,74 \times 10^{-5} + \log \left(\frac{0,10}{0,09} \right);$$

Award mark for using

concentrations and not number of moles;

$$4,759 - 0,2487 = 5,00 \quad 4,51 \quad [1]$$

Change in pH = 5.00 - 4.93 = 0.08

[1]
[Total: 12]

- 4 (a) - melting point increases from Li to C, sharply decreases from C to N and steadily decreases from N to Ne; AW *graph* [1]
 - from Li to Be atoms are held tightly in the metallic lattice; AW *increase in delocalised electrons* [1]
 - a sudden jump when it comes to carbon is due to the giant molecular structure in which strong covalent bonds exist between atoms; AW *strong covalent bonds* [1]
 - N to Ne are simple molecular held by weak van de Waal's forces; AW *charge a bit* [1]
 - a sudden decrease for neon is due to the monoatomic nature of the element; AW [1]

- (b) - (big difference in electro negativity leads to ionic bonding, smaller covalent; [1]
 - sodium oxide ionic lattice (with strong forces between ions; [1]
 - phosphorus oxide simple molecular with weak forces between molecules; [1]

- (c) (i) 1. White flame/white solid; [1]
 2. Spontaneous combustion /burning; AW; *white solid burns with flame a white flame* [1]

- (ii) $Mg + \frac{1}{2}O_2 \rightarrow MgO$; [1]

- $4P + 5O_2 \rightarrow P_4O_{10}$; *BOs / P₄O₆ / P₂O₃* [1]

[Total: 12]

- 5 (a) (i) kills bacteria/prevent bacterial diseases; [1]

- (ii) chlorine is toxic; $Cl_2 + H_2O \rightarrow Cl^- + ClO^- + 2H^+$ [1]

- (iii) $Cl_2 + H_2O \rightarrow HCl + HClO$; (A) ionic [1]

- (iv) bleach, pvc, solvents, photographic paper (any two) *disinfectant* [2]

- (b) HCl; Bond energies H - I - 299 [1]

- H - Cl - 431; [1]

HI bond weaker, breaks easily releasing a higher concentration of H^+ , hence has low pH/AW; [1]

- (c) $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$; (A) ionic *reaction* [1]
 0 -1 +1; [1]

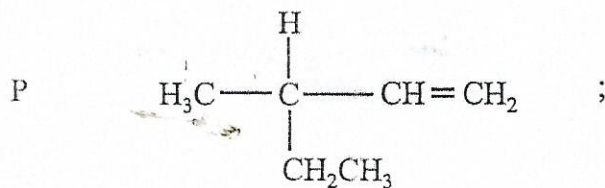
- $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$; [1]

- 0 -1 +5; [1]

/Accept ionic equations;

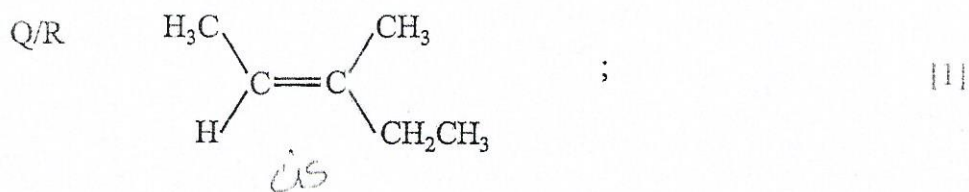
[Total: 12]

6 (a) (i)



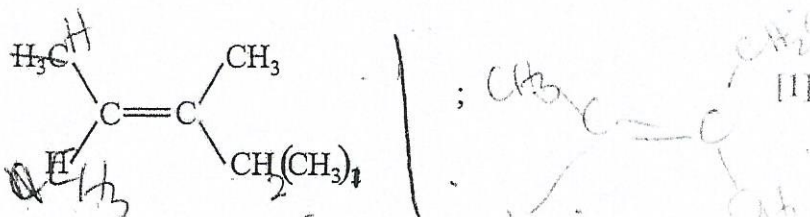
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(ii)



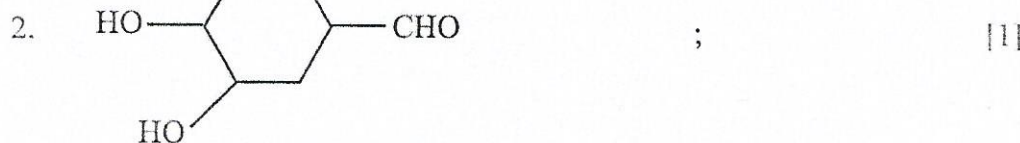
[1]

(iii)



[1]

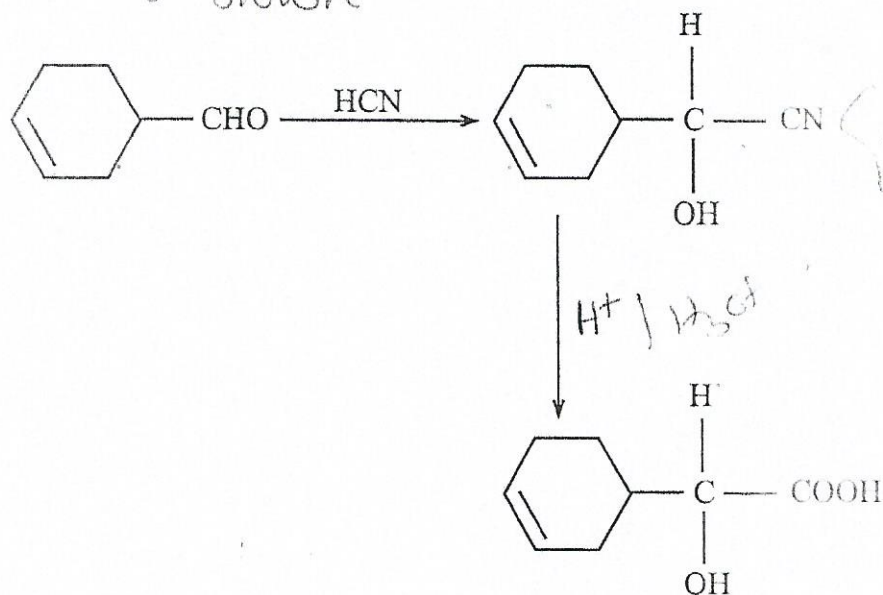
(b) (i)

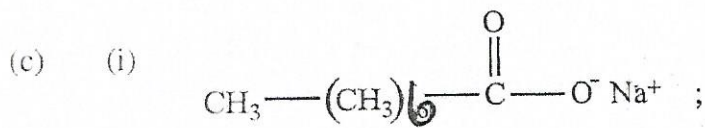


(ii) 1. Red-brown precipitate; [1]

2. purple KMnO_4 turns green; *decolourise to brown* [1]

(iii) [1]





(ii) triester is not soluble products are soluble; *partly soluble*

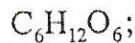
[Total: 12]

7

(a)

(i)

1.



(R)

are

11) - 2)

Structural isomers;

(ii)

can rotate plane polarised light;

all (glucose, fructose and gluconic) acid are chiral/optically active;

(iii)

acidic/low pH/low water content;

can dehydrate bacteria/ *(Low pH)* $\text{H}_2\text{O}_{2(aq)}$ present inhibits growth of bacteria;

(b)

(i)

6;

(ii)

androstenol reacts with Na metal to;

produce $\text{H}_{2(aq)}$;

(iii)

reduction *(using)* LiAlH_4 ;

Reagent 13
oxidations for base 11

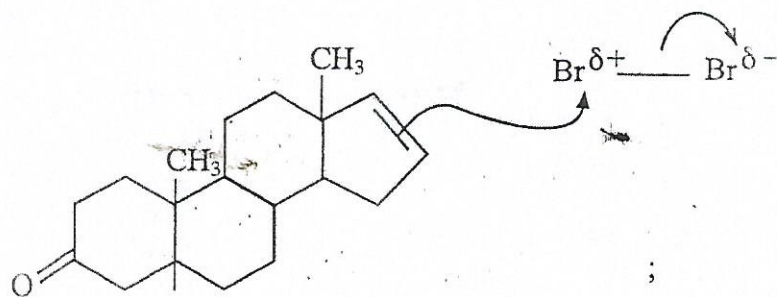
- using ketone

- PCl5

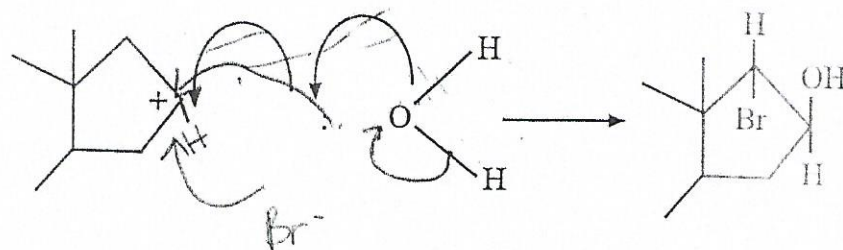
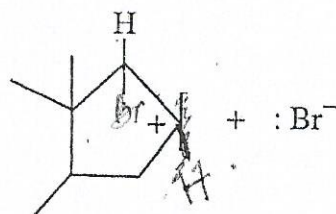
- 2,4 DNP

- Esterification

(iv)



[1]

[1]
[Total: 12]

8

(a)

(i)

1. Addition – elimination / condensation;

[1]

2. electrophilic substitution; ^(A)nitration

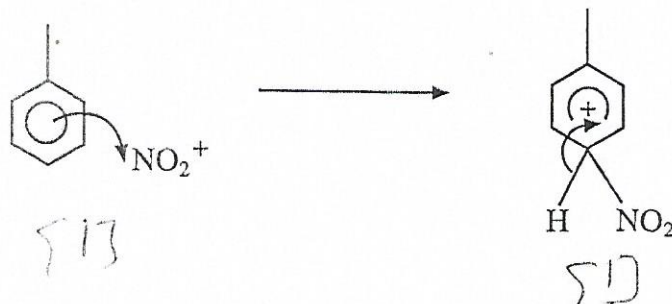
[1]

(ii) NaOH, reflux; (reject acid)

[1]

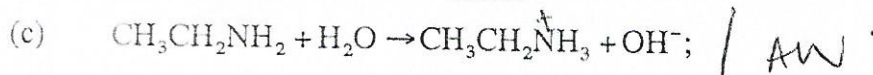
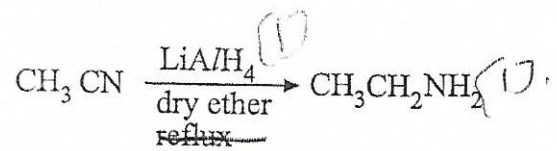
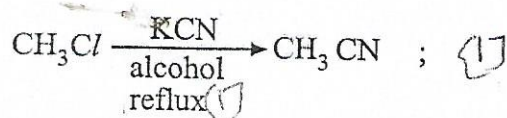
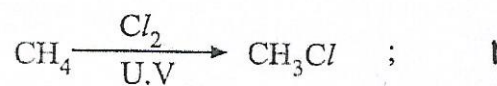
(iii) $\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{H}_3\text{O}^+ + 2\text{HSO}_4^-$;

[1]



[2]

(b)



Salt formation

[Total: 12]